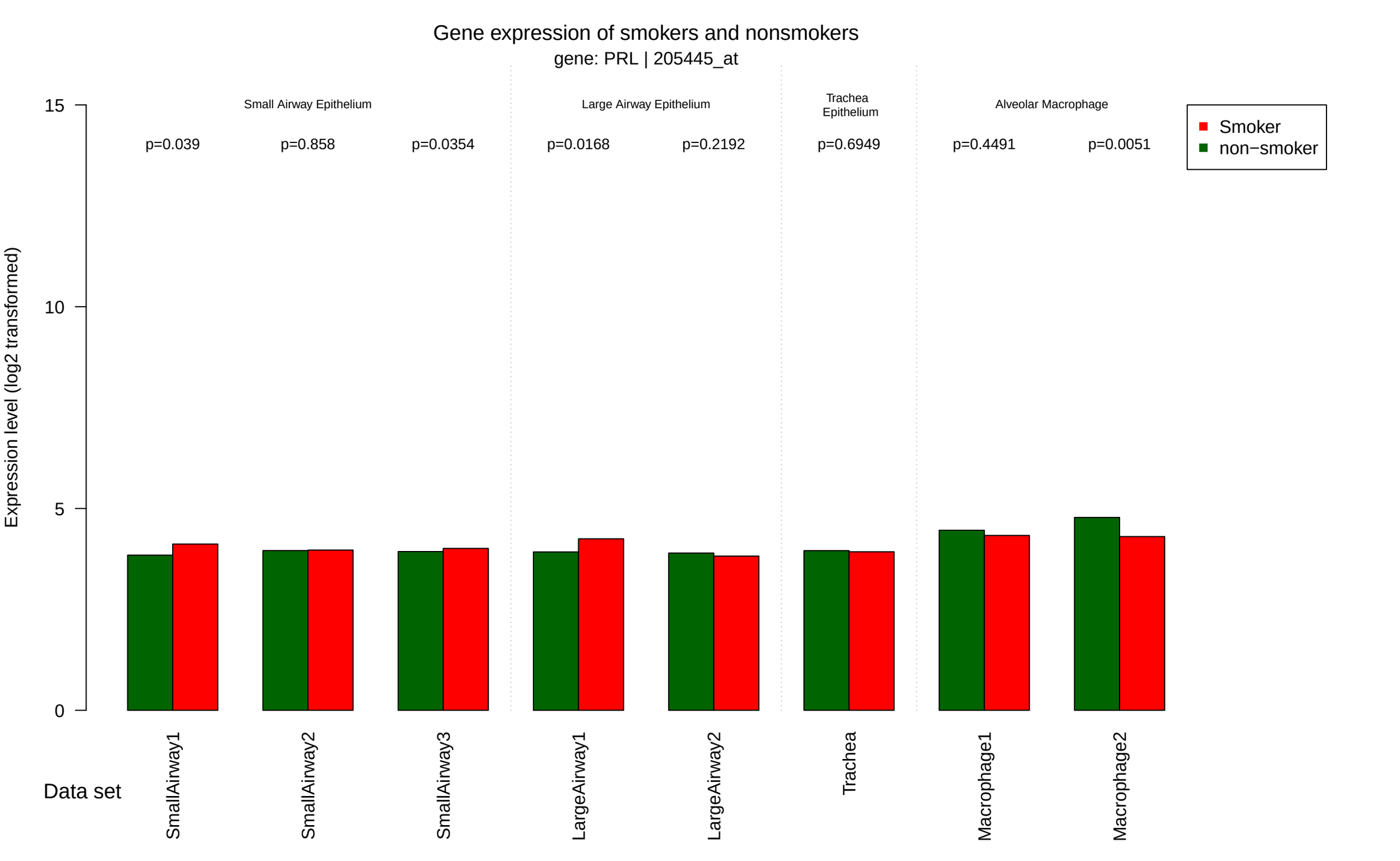
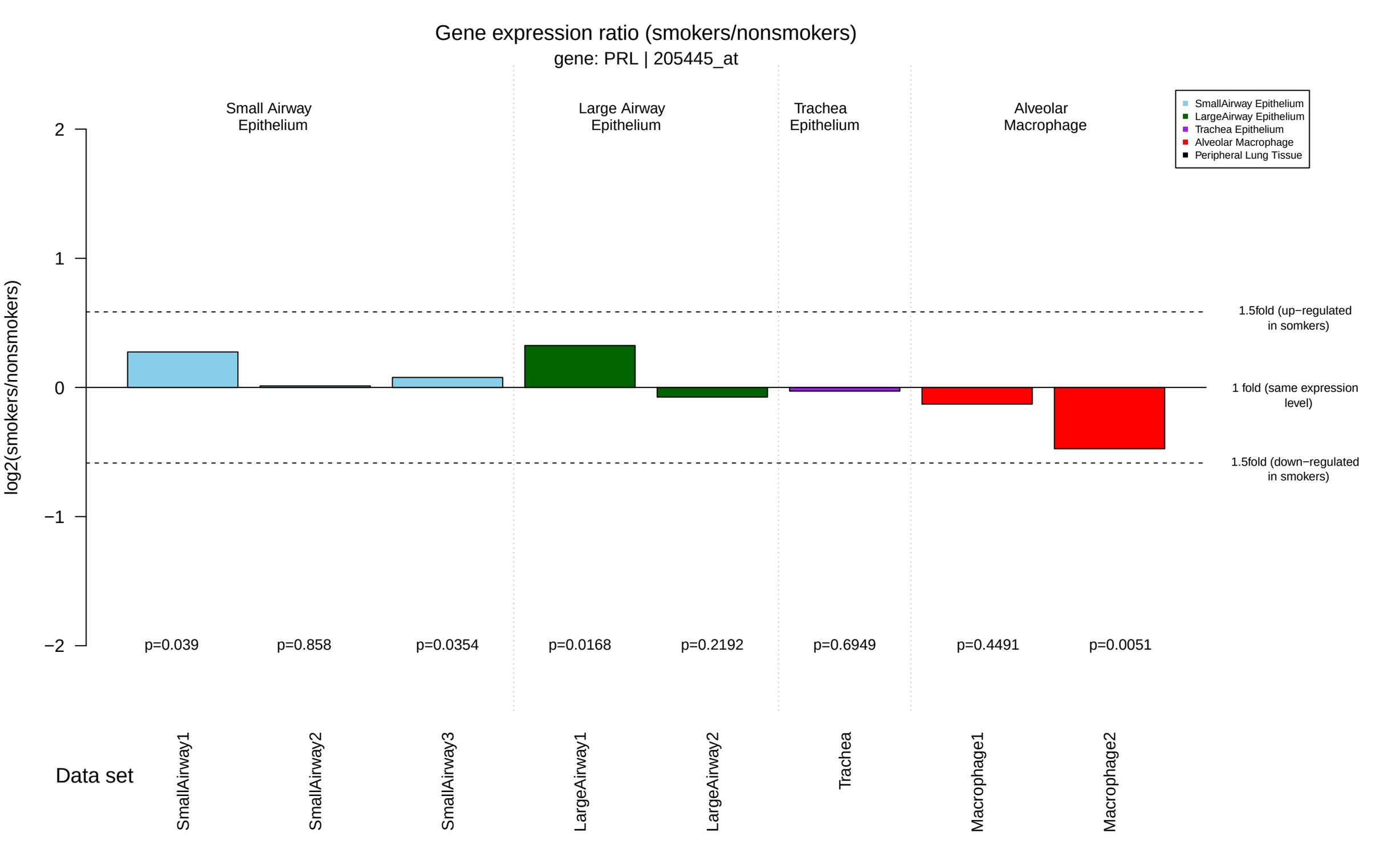
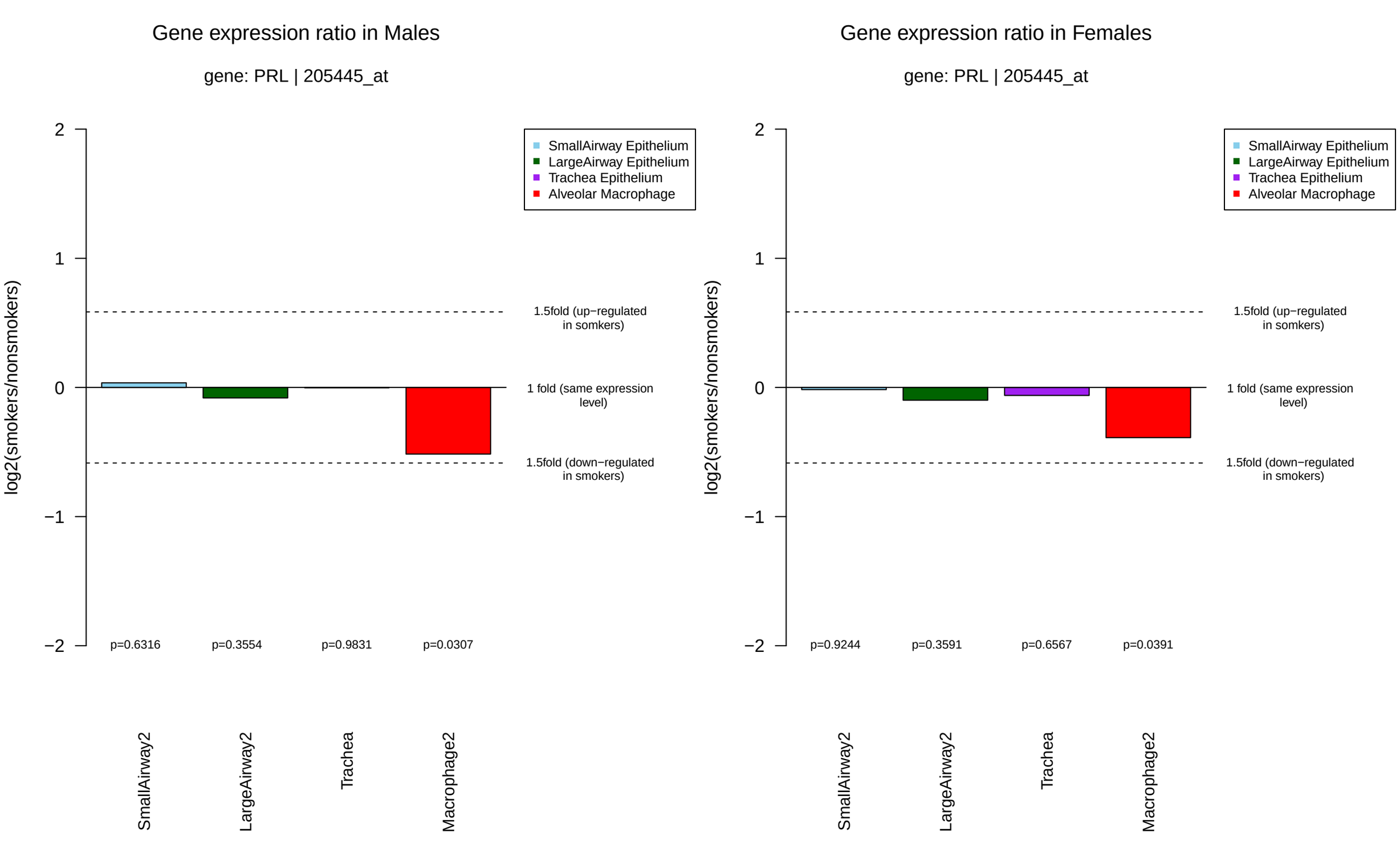
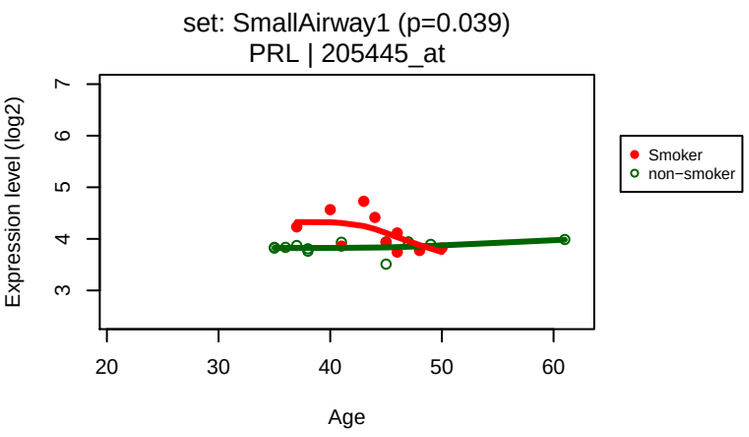
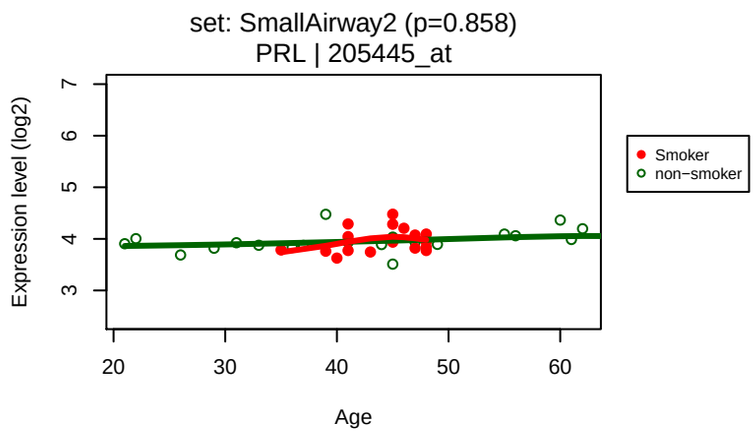
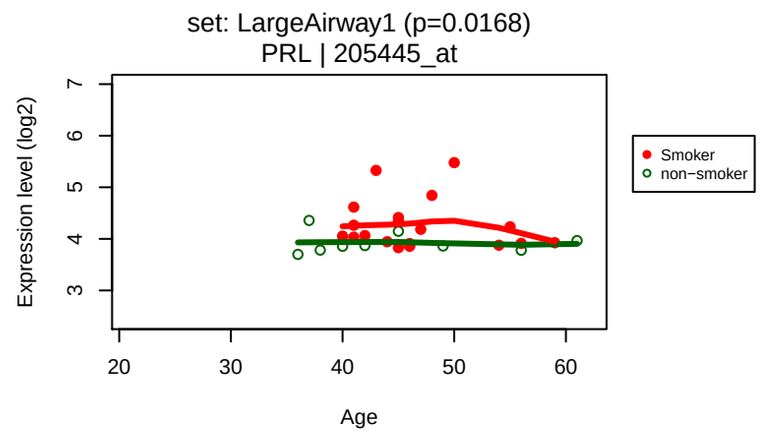

Gene expression of smokers and nonsmokers
gene: PRL | 205445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.039
p=0.858
p=0.0354
p=0.0168
p=0.2192
p=0.6949
p=0.4491
p=0.0051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

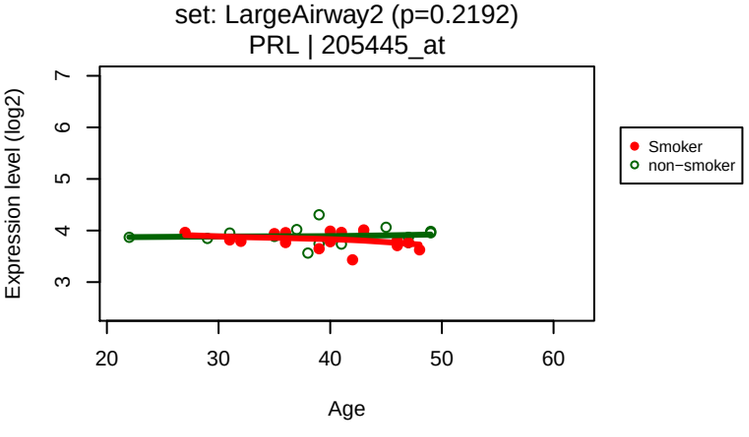

Gene expression ratio (smokers/nonsmokers)
gene: PRL | 205445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.039
p=0.858
p=0.0354
p=0.0168
p=0.2192
p=0.6949
p=0.4491
p=0.0051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

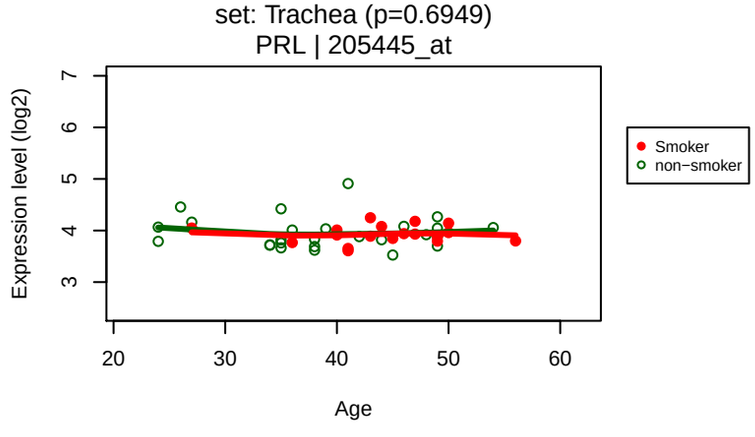

Gene expression ratio in Males
gene: PRL | 205445_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6316
p=0.3554
p=0.9831
p=0.0307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRL | 205445_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9244
p=0.3591
p=0.6567
p=0.0391
SmallAirway2
LargeAirway2
Trachea
Macrophage2

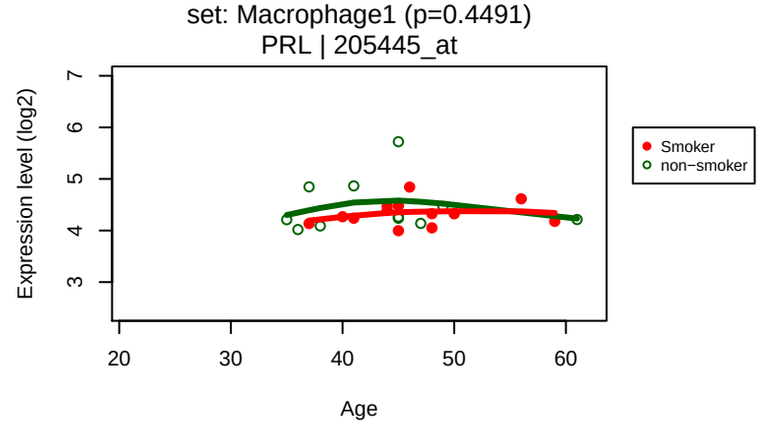

set: SmallAirway1 (p=0.039)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker

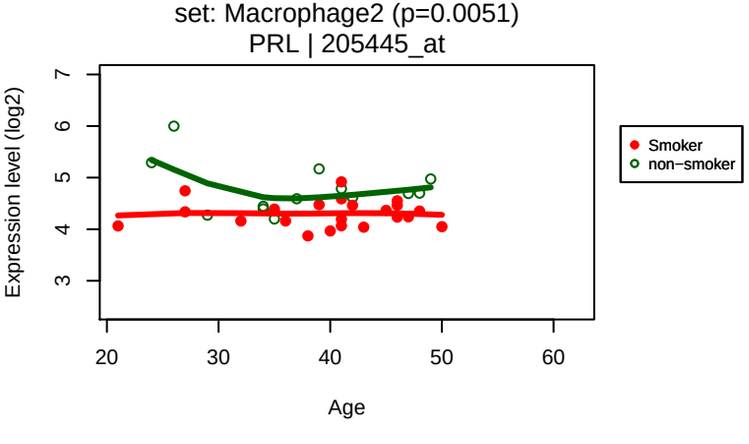

set: SmallAirway2 (p=0.858)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker

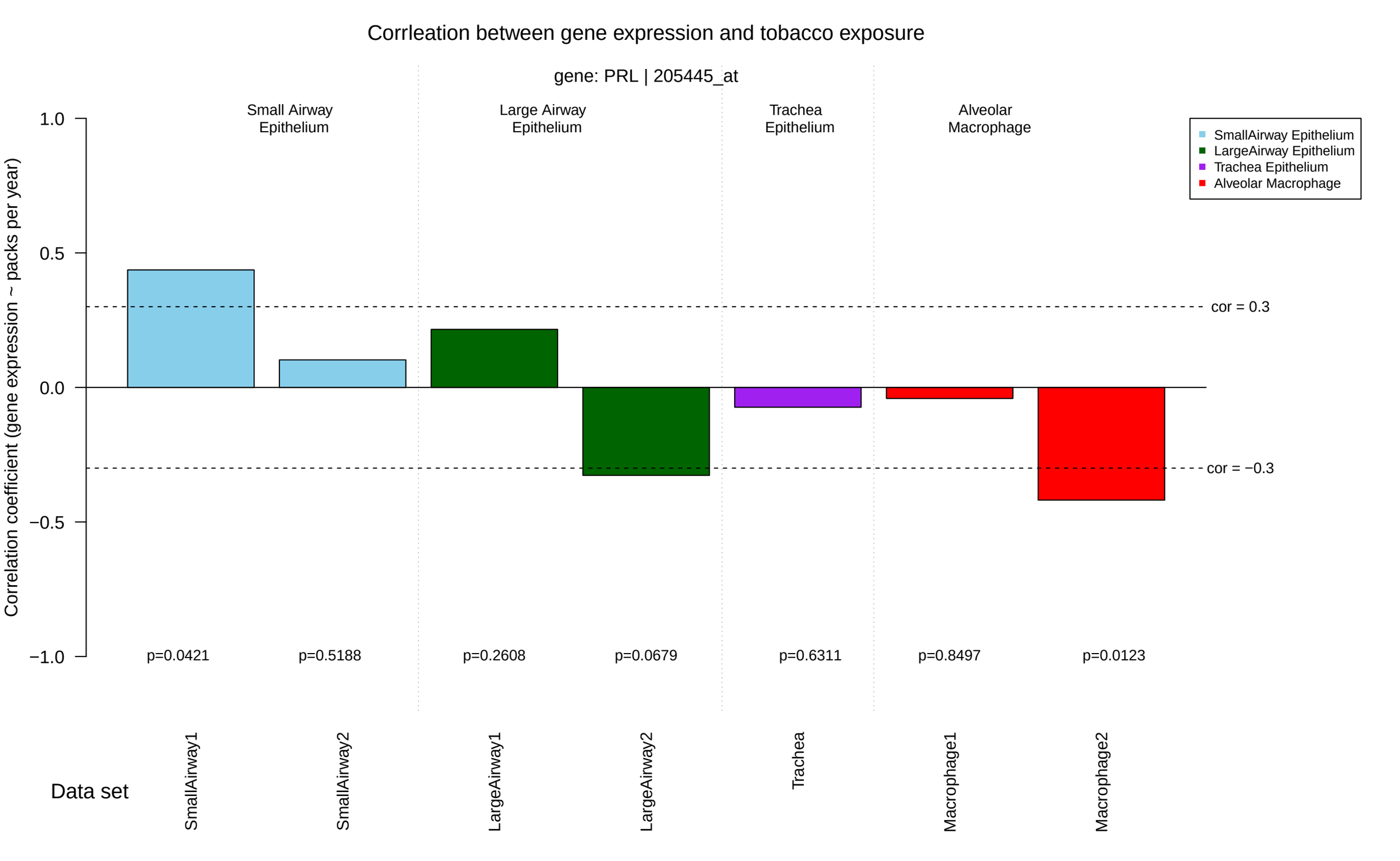

set: LargeAirway1 (p=0.0168)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2192)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6949)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4491)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0051)
PRL | 205445_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRL | 205445_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0421
p=0.5188
p=0.2608
p=0.0679
p=0.6311
p=0.8497
p=0.0123
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2